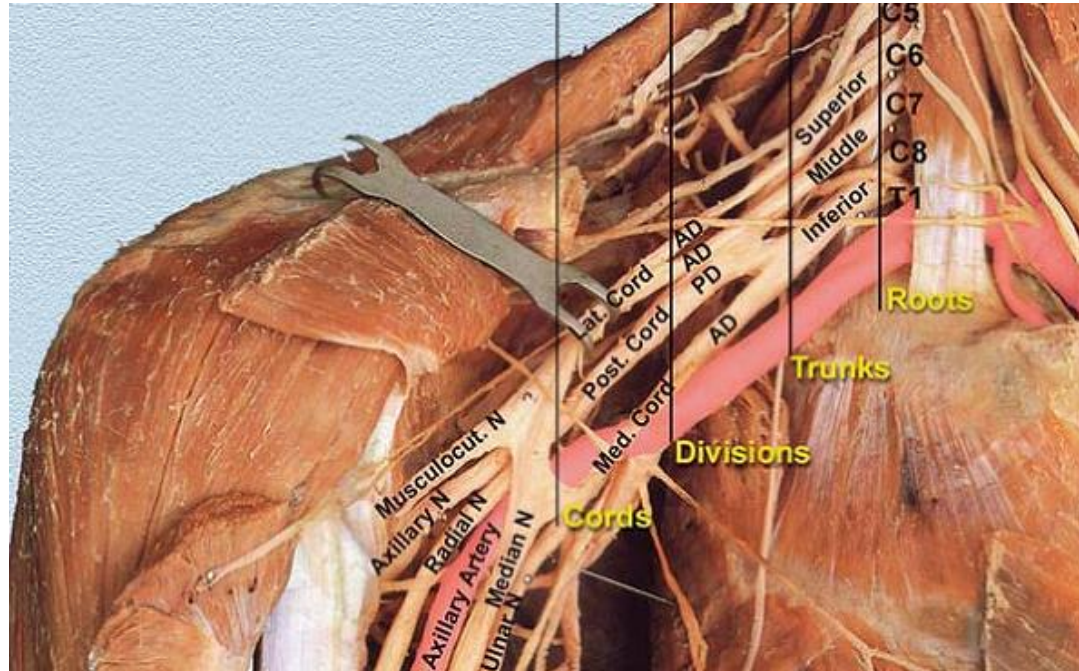


Brachial Plexus Self-Study



IS-MSN, Fall 2024

Mark F. Teaford, Ph.D.

Andrea Taylor, Ph.D.

Theo Smith, Ph.D.

Amanda Smith, Ph.D.

When you have completed this tutorial, you should be able to:

- 1) Discuss and explain the formation of the brachial plexus.
- 2) Identify and discuss the major components of the brachial plexus and their branches.
- 3) Know the spinal cord contributions that contribute to each of the major components of the brachial plexus (roots, trunks, divisions, cords, and terminal branches) and to the named nerve branches.
- 4) Be able to draw and label the major components of the brachial plexus and their branches.
- 5) Be able to discuss and explain the three-dimensional relationships of the brachial plexus, particularly in relationship to muscle-tendon units, major arteries, and adjacent bony structures. Consider where components of the brachial plexus could be mechanically compressed.
- 6) Be able to address specific questions (in red) that you will encounter throughout this tutorial.

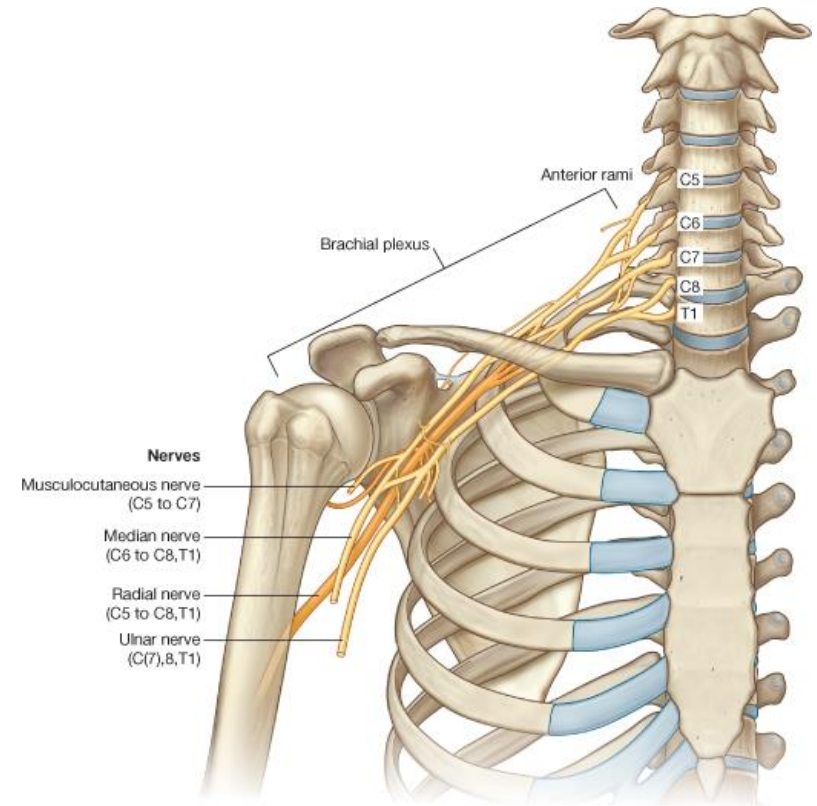
General Overview:

An anatomical plexus is a network of nerves.

The brachial plexus is a network of **somatic nerves** formed by the union of **ventral rami C5-T1**, with variable contributions from C4 and T2. Remember that ***all*** somatic plexuses are formed by ventral rami of spinal nerves.

The brachial plexus is responsible for innervating most of the upper limb.

The brachial plexus originates in the cervical region. As the ventral rami forming the plexus join and separate, the plexus runs distal-laterally, inferior to the clavicle, to enter the axilla. Branches of the plexus innervate the superficial back and extrinsic shoulder regions, the scapular, deltoid and pectoral regions as well as the arm, forearm and hand.

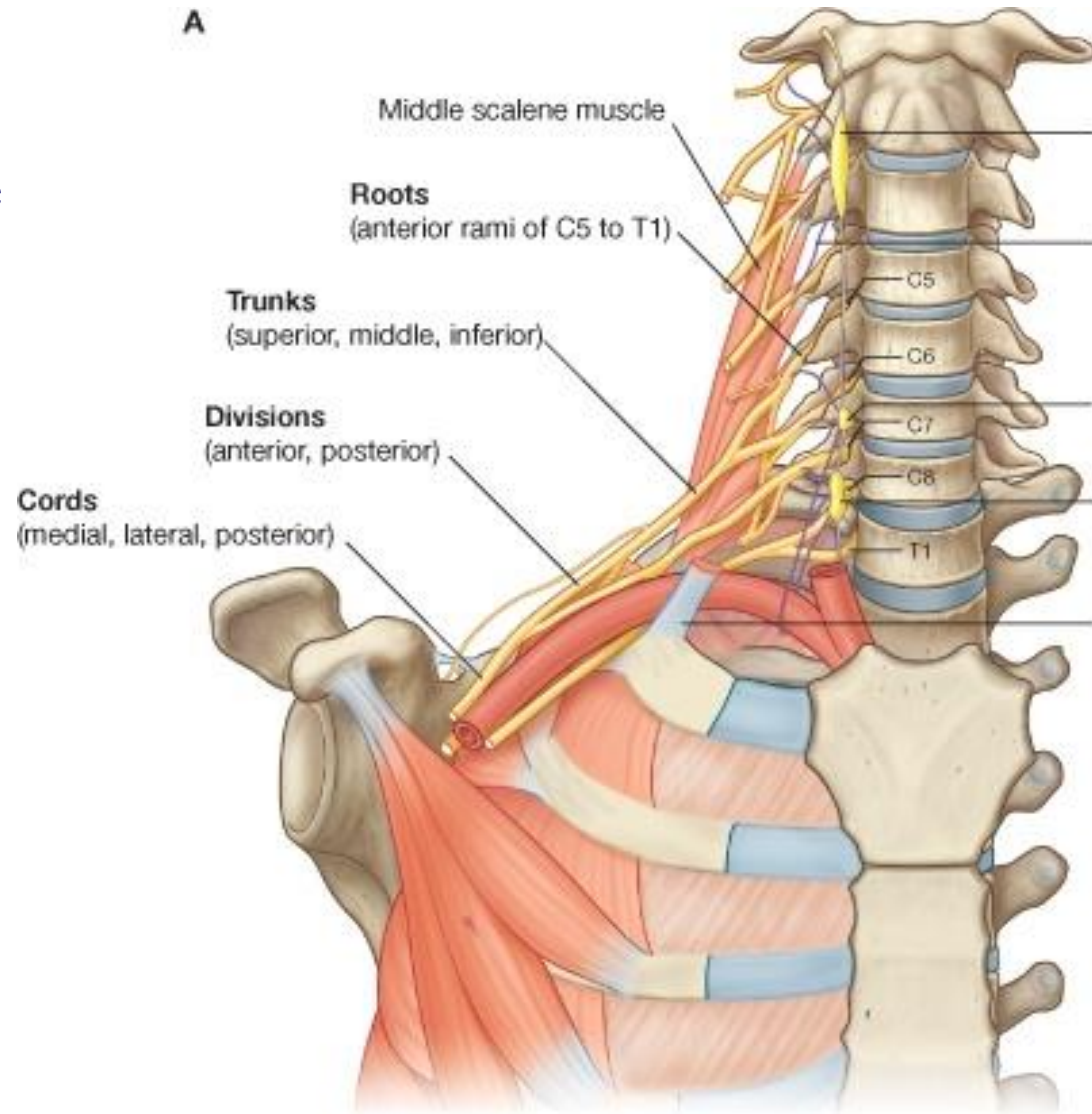


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There are 5 major components to the plexus:

- Roots** (which are the ventral rami of the spinal nerves; don't confuse the roots of the plexus with ventral or dorsal roots of *spinal nerves*),
- Trunks**, **Divisions**, **Cords**, and **Terminal Nerves** (also known as **terminal branches**).

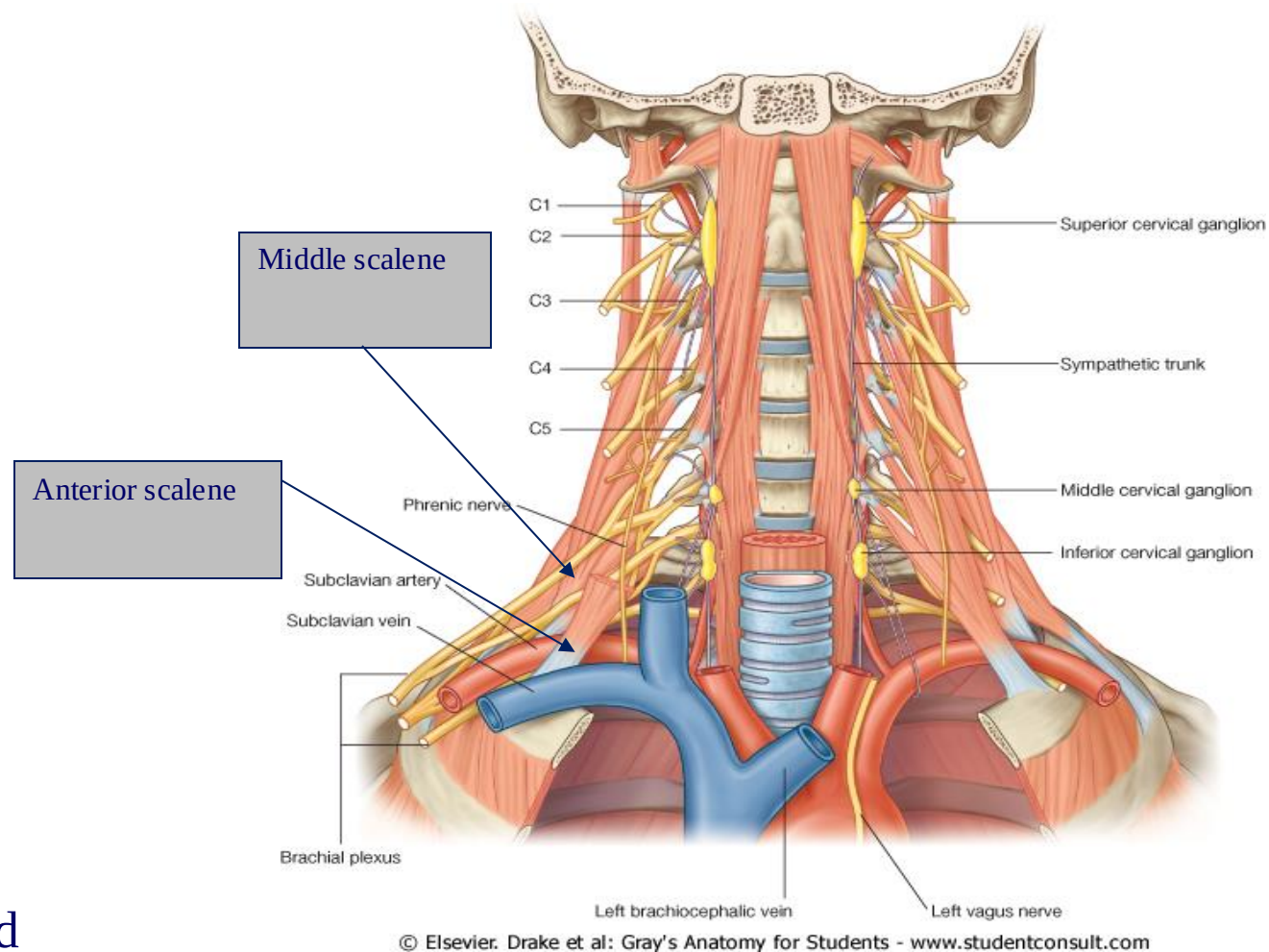
Note that in this figure, the terminal nerves or branches are not present or labeled. These are clearly depicted, labeled and discussed on Slide #11.



Roots and their branches:

The ventral rami that form the '**roots**' of the brachial plexus derive from spinal nerves C5-T1. These are ventral rami of spinal nerves so they carry both **somatomotor** and **somatosensory** fibers.

The ventral rami (roots) of the plexus emerge in the neck, sandwiched between the **anterior** and **middle scalene** mm.

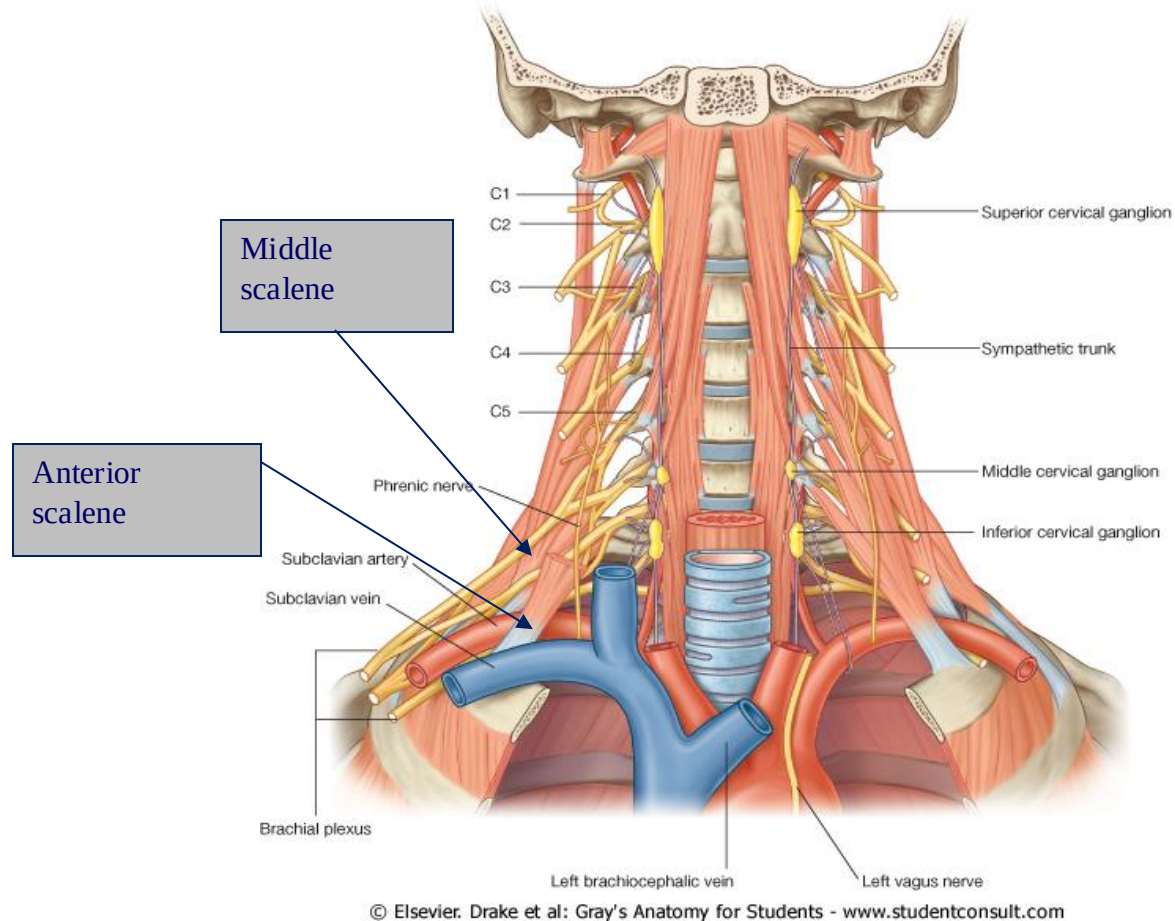


Anterior View

Roots and their branches:

The roots of the plexus give off the following (note that the spinal nerve contributions to each of the named peripheral nerves are given in parentheses):

- **dorsal scapular nerve** (C5), which supplies the rhomboids and levator scapulae muscles
- **long thoracic nerve** (C5-C7), which supplies serratus anterior
- contributions to the **phrenic nerve** (a somatic nerve that supplies the diaphragm; C3-C5).



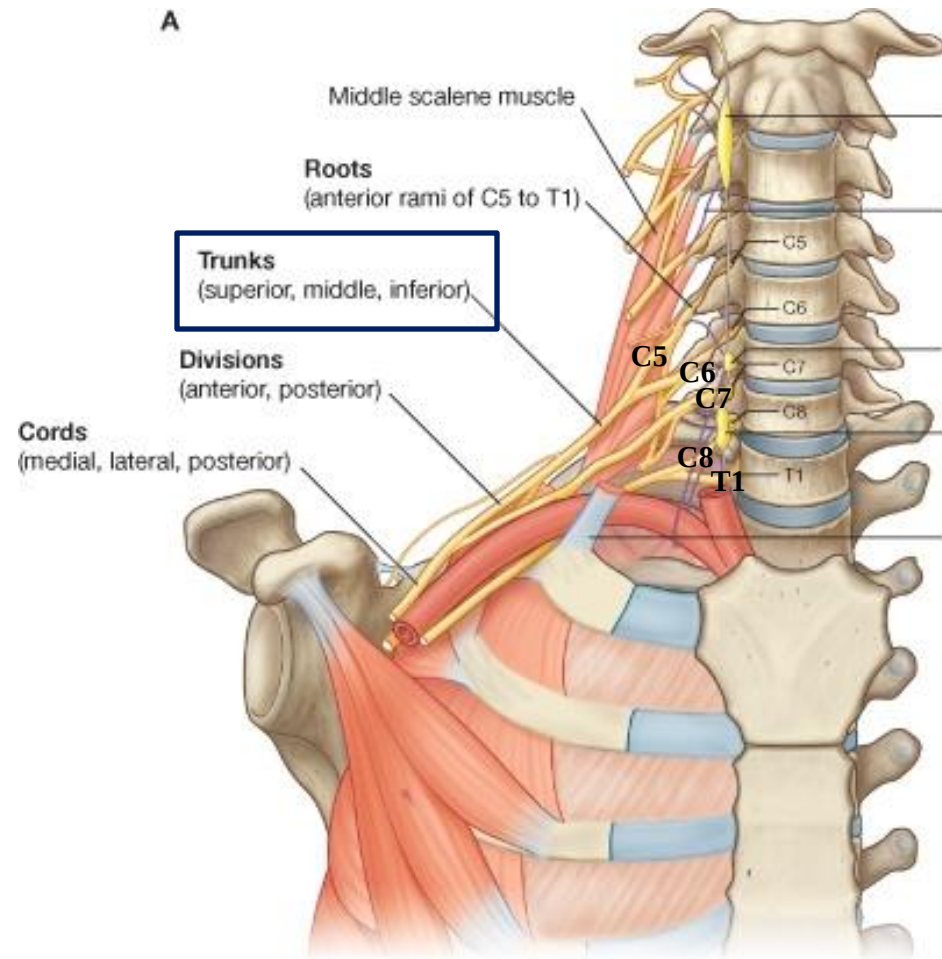
Trunks and their branches:

The roots of the plexus unite to form three trunks, the **upper** (superior), **middle**, and **lower** (inferior) trunks. The trunks are situated proximal to the clavicle.

The upper trunk is formed by the union of ventral rami C5-C6. The lower trunk is formed by the union of ventral rami C8-T1. The middle trunk is formed simply by the ventral ramus of C7.

Only one trunk – the upper trunk – gives rise to any nerve branches. These are

- *suprascapular nerve*, which supplies the supraspinatus and infraspinatus mm.
- *nerve to subclavius* (a small muscle situated between the inferior clavicle and 1st rib).

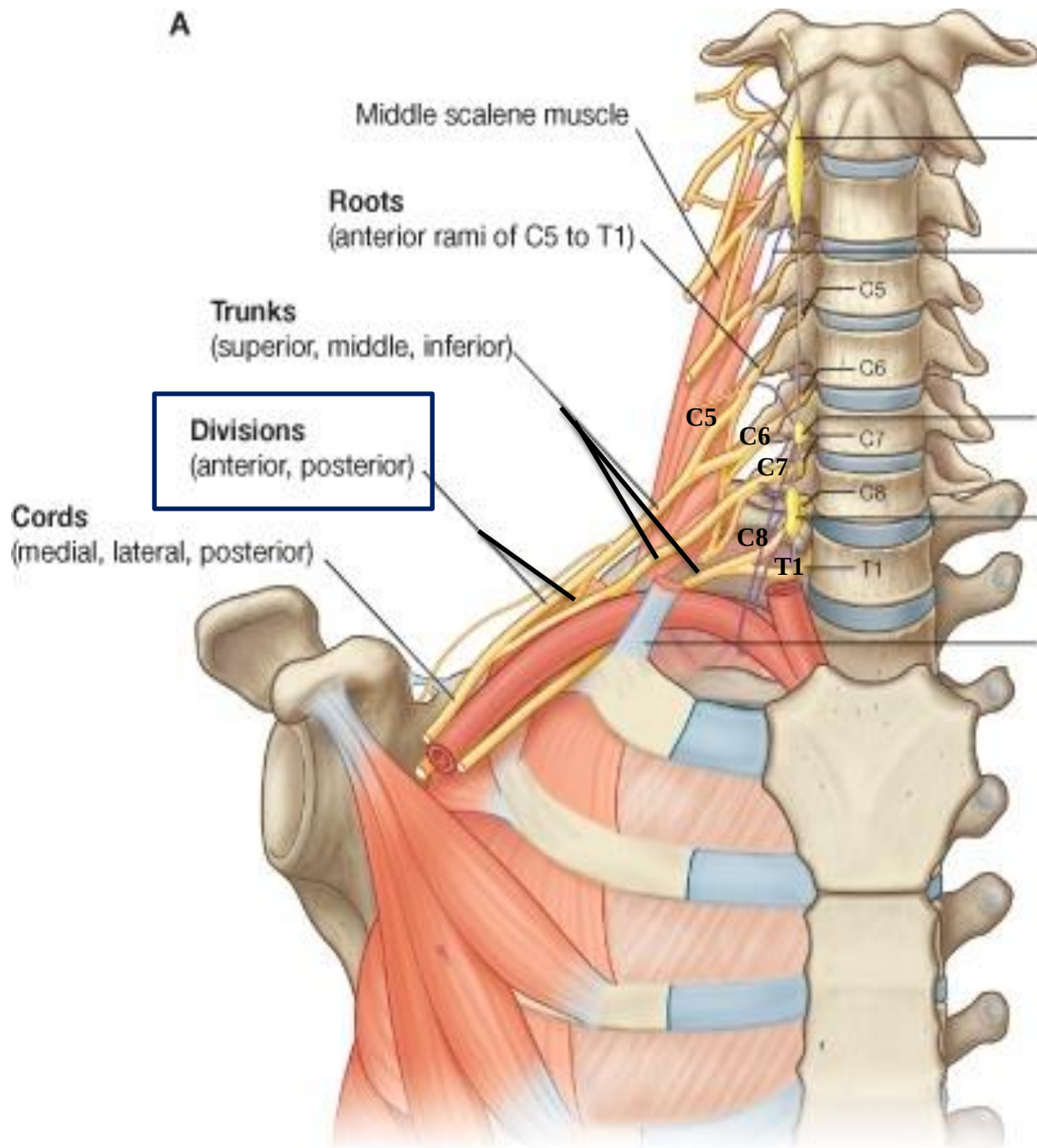


Divisions:

Each of the three trunks splits into an **anterior division** and a **posterior division**. These divisions may be positioned proximal to the clavicle, or between the clavicle and 1st rib; this is variable.

In this figure, only the anterior and posterior divisions of the upper trunk are labeled; the posterior divisions of the other trunks are deep and not visible in this figure.

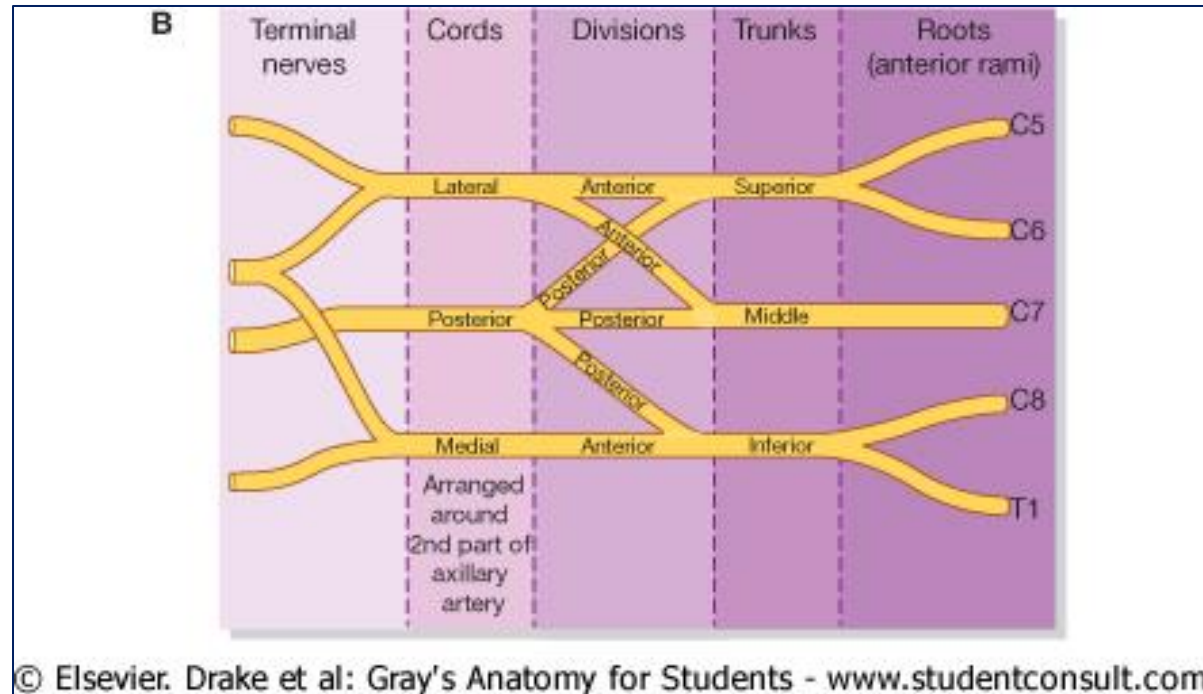
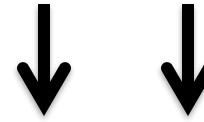
No branches are formed off of the divisions of the plexus.



Divisions and Cords:

The contributions of the divisions to the cords and of the divisions to the continuation of the plexus are more easily visualized in this schematic:

- anterior divisions from the upper and middle trunks unite to form the lateral cord;
- the anterior division of the inferior trunk forms the medial cord.
- The posterior divisions from all three trunks unite to form the posterior cord.



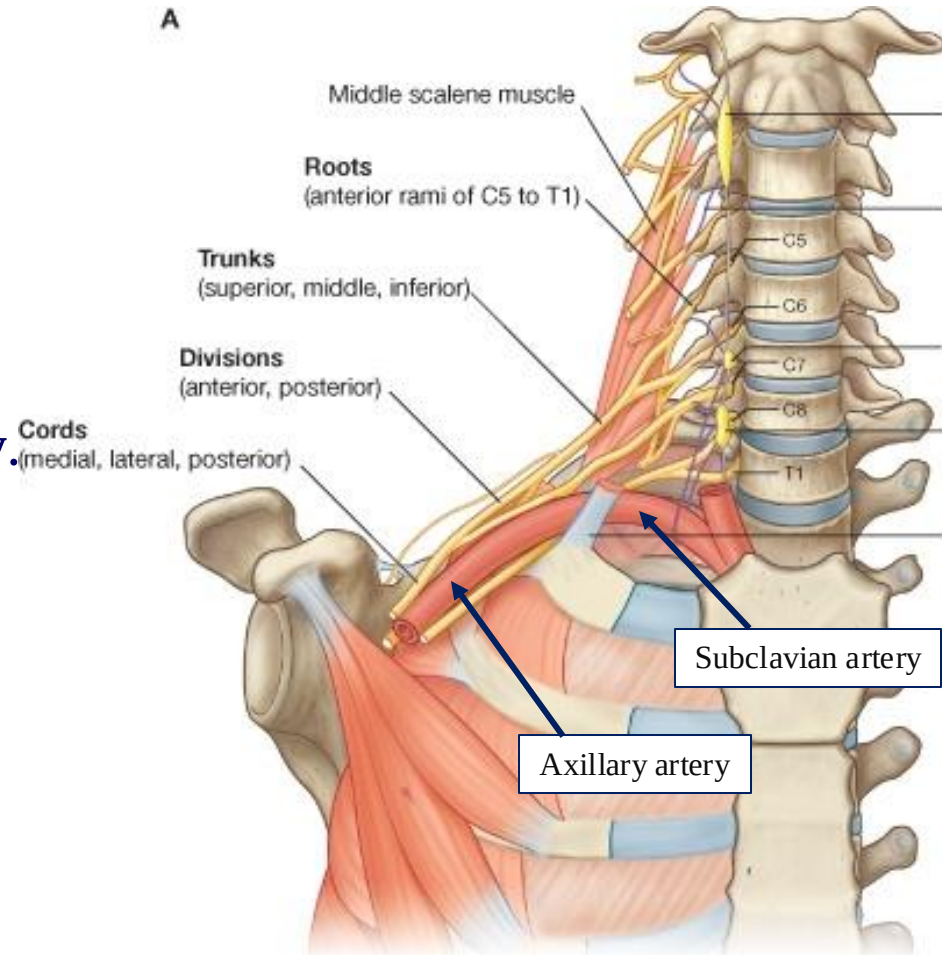
Cords:

The three cords – lateral, medial, and posterior – are so named according to their position relative to the **axillary artery**, the major artery to the upper limb.

Note that the axillary artery is a continuation of the **subclavian artery**. The subclavian artery changes its name to axillary artery when the subclavian artery exits from deep to the clavicle to reach the 1st rib.

The cords generally run deep to the clavicle and deep to the tendon of insertion of the pectoralis minor.

Q: What spinal cord levels contribute to each cord? The schematic figure from the previous slide will be helpful in answering this question.



Prior to giving off the 5 terminal nerves of the plexus, the cords provide origin for 7 named peripheral nerves:

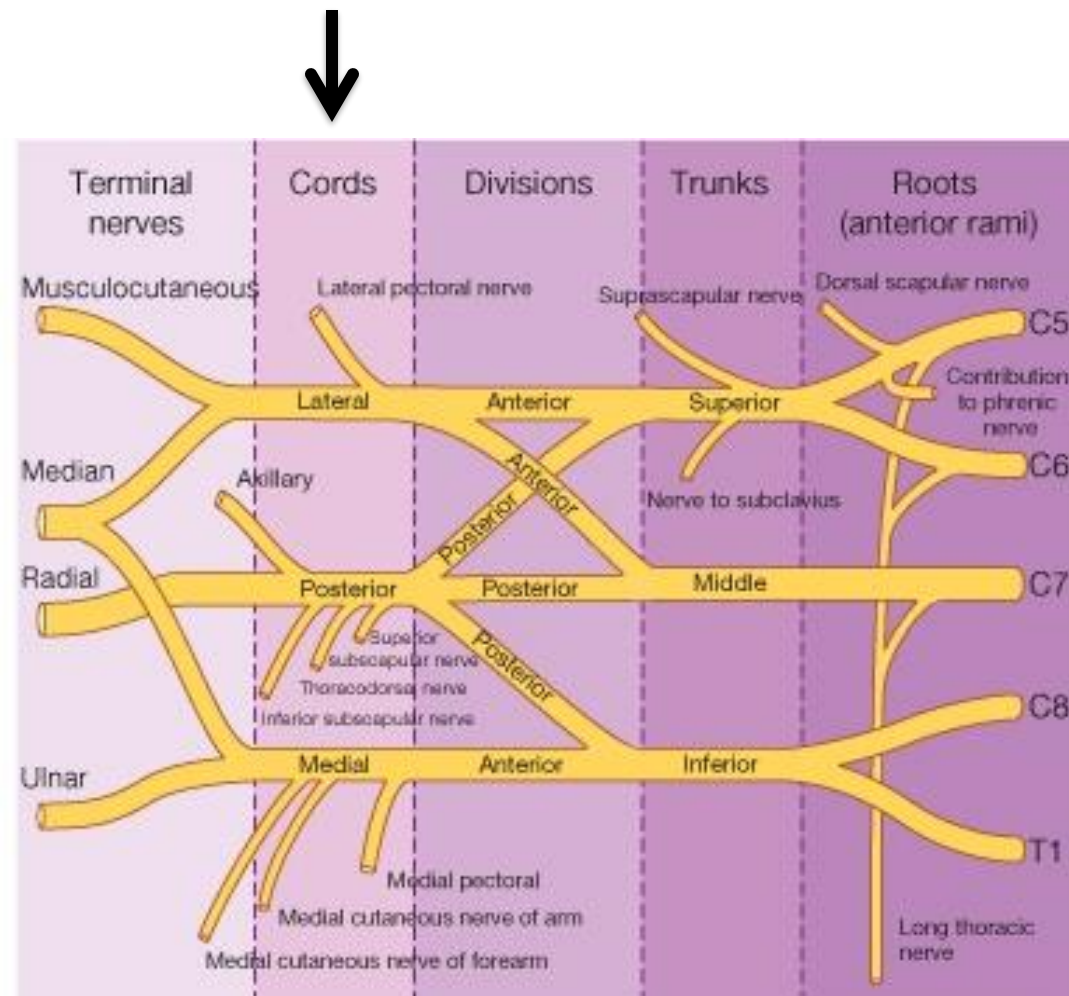
The lateral cord gives off one nerve, the **lateral pectoral nerve**.

The medial cord gives off three nerves: **medial pectoral nerve** and two nerves that are solely cutaneous nerves: the **medial brachial cutaneous nerve** and **medial antebrachial cutaneous nerve**.

The branches directly off of the posterior cord are the **upper (superior) subscapular nerve, thoracodorsal nerve, and lower (inferior) subscapular nerve**.

Q: What muscles are innervated by these last three nerves?

Q: What are the spinal cord contributions to each of these 7 nerves?

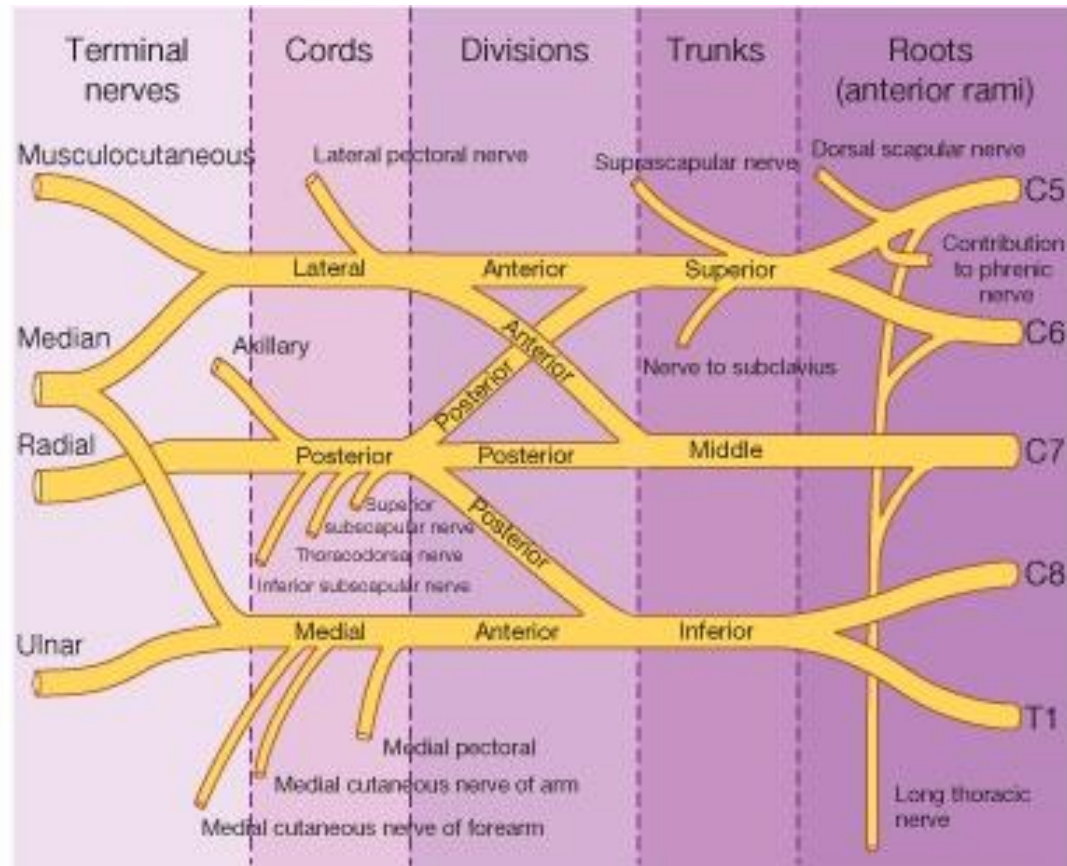


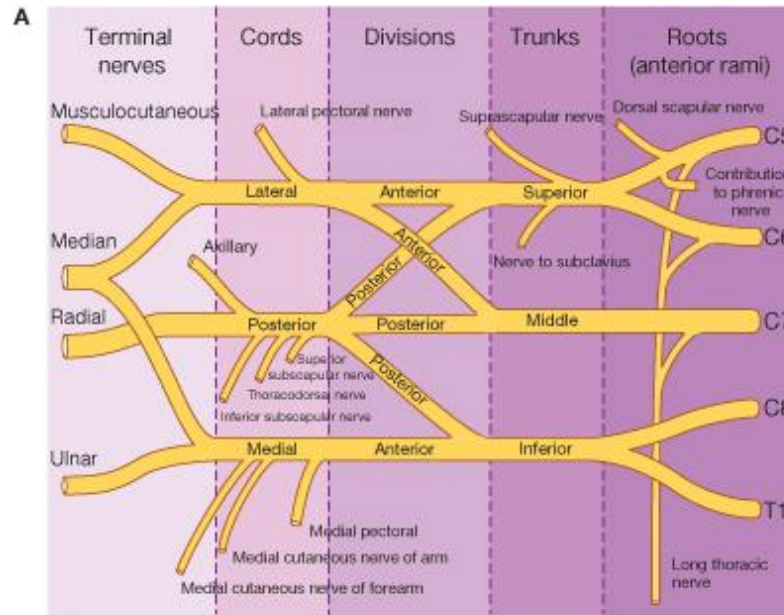
[lateral pectoral = C5-7; medial pectoral = C8-T1; medial cutaneous nerves = C8-T1; upper & lower subscapular nerves = C5-6; thoracodorsal nerve = C6-8]

Terminal nerves:

There are five terminal nerves: **musculocutaneous, median, ulnar, radial and axillary nn.** These terminal nerves provide the innervation to the remainder of the upper limb.

You should know i) the specific cords of the plexus from which each of these terminal nerves originate; and ii) the spinal cord contributions to each of these terminal nerves.





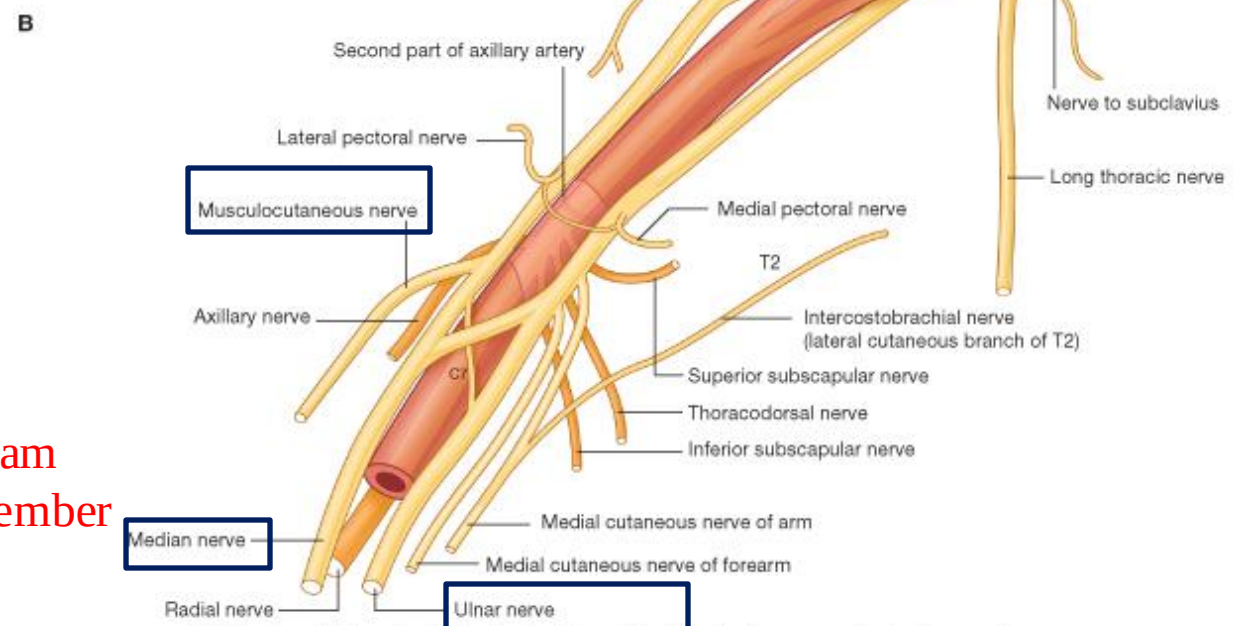
Terminal nerves:

Musculocutaneous nerve derives from the lateral cord and receives contributions from spinal cord levels C5-C7.

Ulnar nerve derives from the medial cord and receives contributions from C8-T1.

Median nerve is formed by contributions from both the lateral and medial cords (C6-T1).

[Refer to the schematic diagram at the top of this slide to remember why these nerves have these spinal cord contributions]

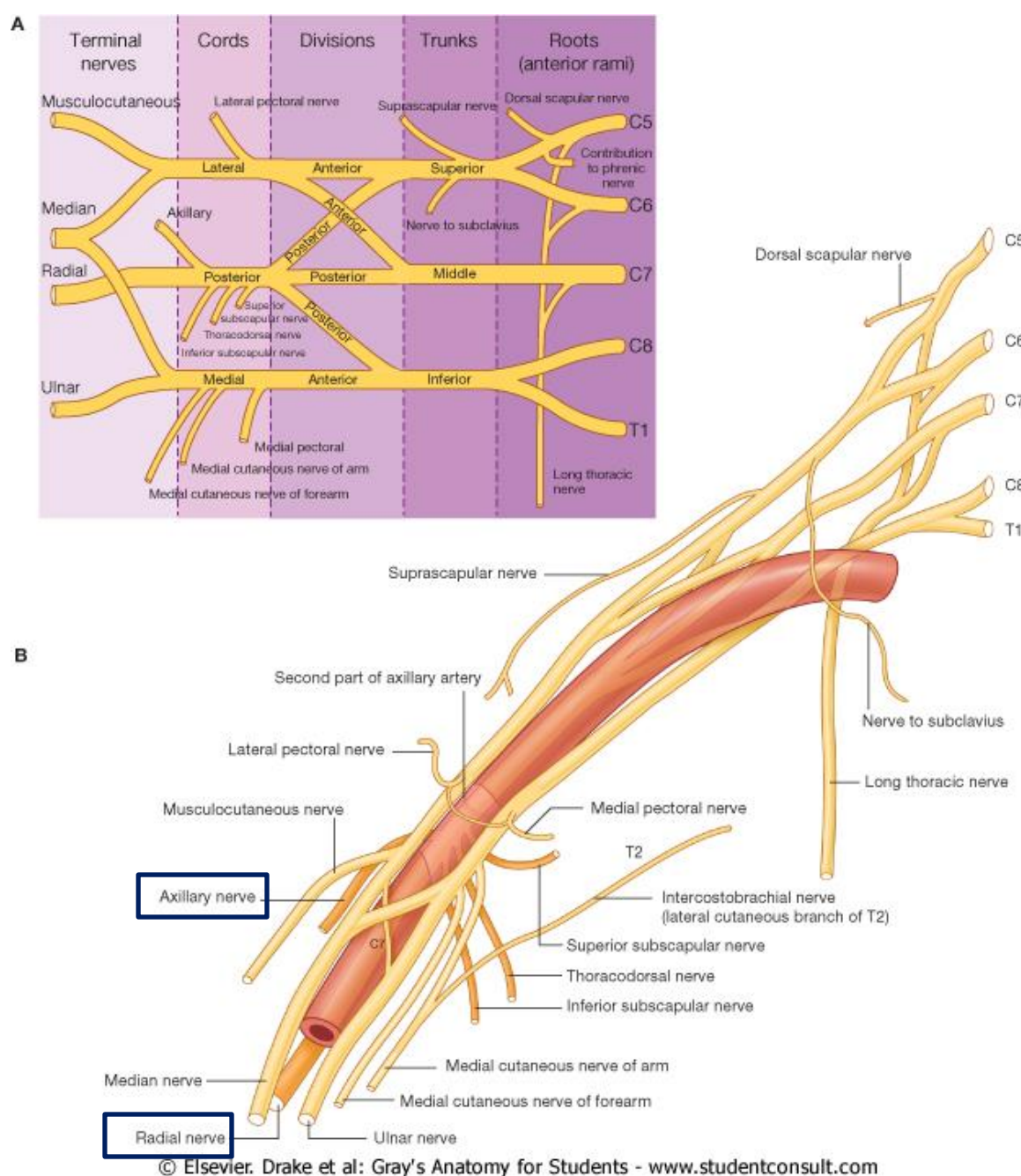


Terminal nerves (Cont'd):

Radial (C5-C8) and **axillary** (C5-6) nerves derive from the posterior cord.

Radial nerve is the largest of the five terminal branches because it is responsible for innervating all of muscles of the extensor compartments of the arm and forearm.

Axillary nerve supplies two muscles: teres minor and deltoid, and then terminates as a cutaneous nerve to serve the skin of the upper lateral arm.



Note that the terminal branches that include musculocutaneous, median and ulnar nerves form the letter 'M' (red dotted lines at right). This 'M' formation is very helpful in lab.

Lastly, be sure to pay attention to the relationship of the 5 terminal branches to the **axillary artery**. Be able to explain the location of these branches relative to this artery. For example, the musculocutaneous nerve would be found traveling lateral to the axillary artery.

